

POLITECNICO DI TORINO

Master's Degree in Electronics



Master's Degree Thesis

Loss analysis of a 4-level, flying capacitor, bridge-less, totem-pole PFC

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Chapter 1

SPICE models Validation

In order to properly analyze more critical components in terms of losses, it is useful to first create a digital model of the device; a validation procedure of the SPICE models of each device has been performed. The following sections describe the procedure and results of this validation procedure.

All simulation results are available [1]

1.1 Substitute slow leg Si FET model validation

This section describes the behavior of the SPICE model of IPQC60T010S7A, compared to the datasheet. [2] All results are intended for $T_j = 25^\circ\text{C}$ unless otherwise noted.

1.1.1 Static characteristics

Op analysis

Parameter	Datasheet(typ)	Simulation	Notes
$V_{(GS)th}$	3.5V	3.75V	$V_{DS} = V_{GS}, I_D = 3.06\text{mA}$ 1.2
I_{DSS}	$8\mu\text{A}$	$7.2\mu\text{A}$	$V_{DS} = 600; V_{GS} = 0, T_j = 25^\circ\text{C}$ 1.1
I_{GSS}	100nA	40nA	$V_{DS} = 0; V_{GS} = 20$ 1.1

Table 1.1: static characteristics comparision

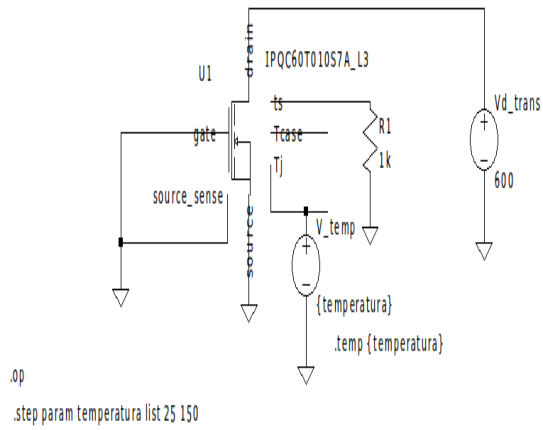


Figure 1.1: zero gate validation circuit

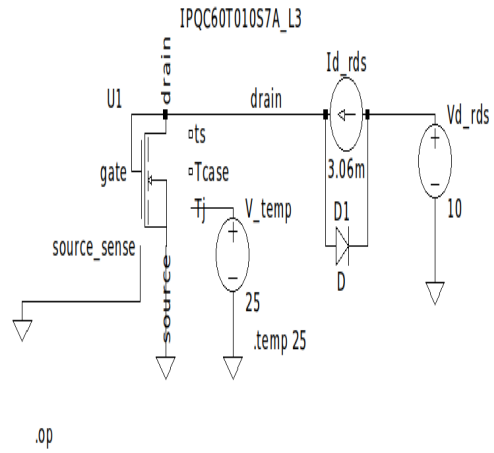


Figure 1.2: Vth validation circuit

Trans charateristics

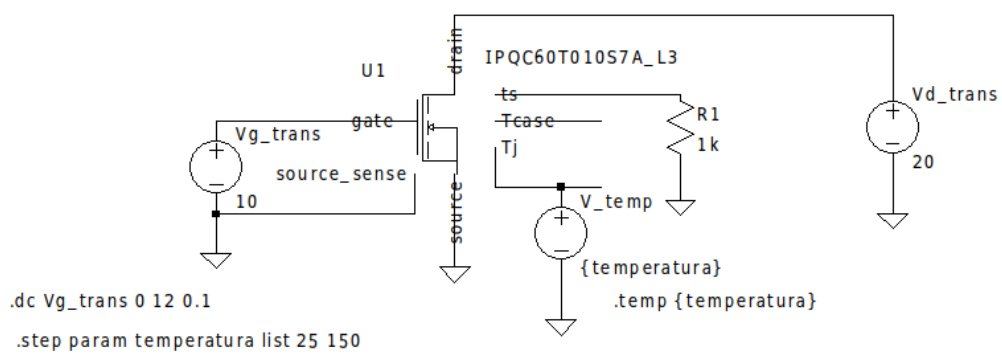


Figure 1.3: Schematic for Trans char validation

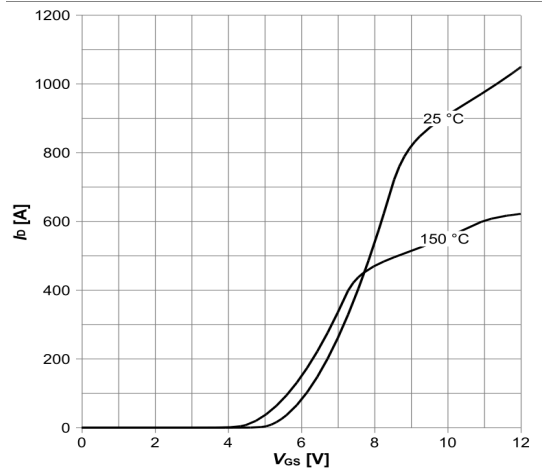


Figure 1.4: Datasheet Trans char

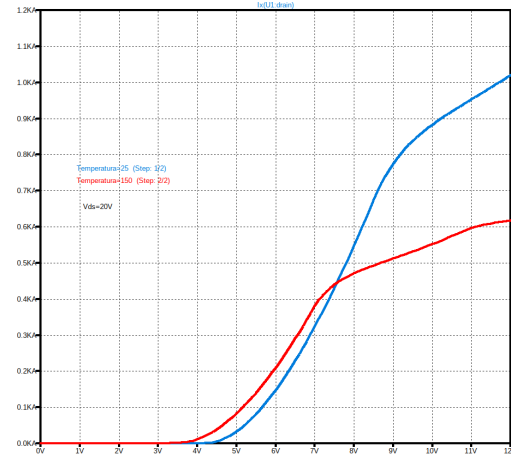


Figure 1.5: SPICE Tran char

$T_j; V_{GS}$	I_D Datasheet	I_D Simulation	Notes
$T_j = 150^\circ\text{C}; V_{GS} = 11\text{V}$	$\approx 600\text{A}$	598.5A	-
$T_j = 25^\circ\text{C}; V_{GS} = 10\text{V}$	$\approx 900\text{A}$	893.1A	-
$T_j = 150^\circ\text{C}; V_{GS} = 3.2\text{V}$	$\approx 0\text{A}$	0.742A	Vth
$T_j = 25^\circ\text{C}; V_{GS} = 4.5$	$\approx 0\text{A}$	0.742A	Vth

Table 1.2: trans characteristics comparision

Output charateristics

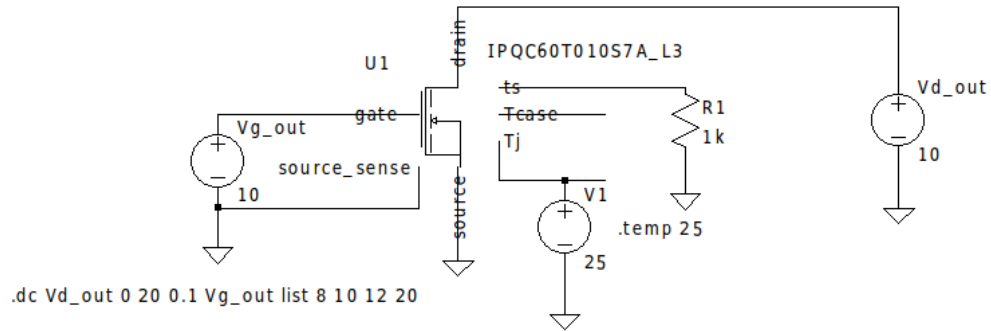


Figure 1.6: Schematic for Si I-V validation

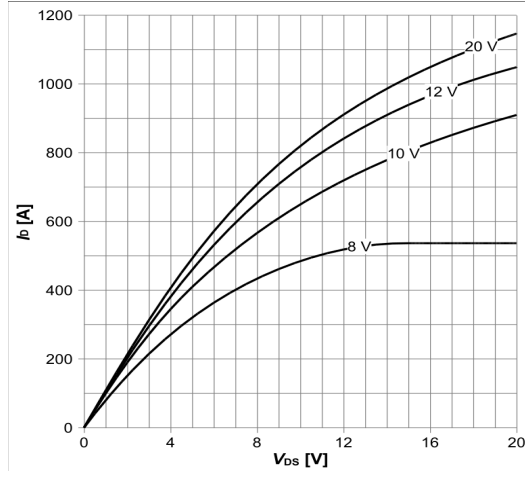


Figure 1.7: Datasheet output char @25°C

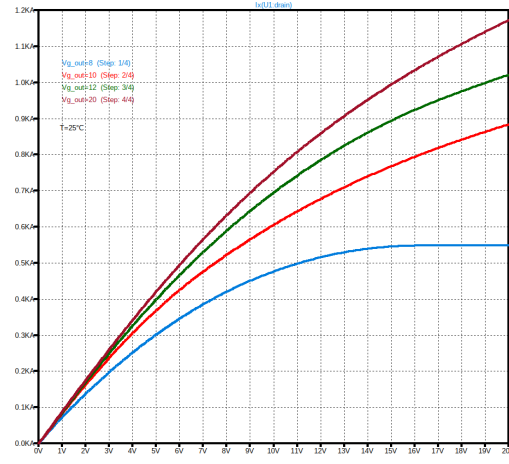


Figure 1.8: SPICE output char @25°C

$V_{DS}; V_{GS}$	I_D Datasheet	I_D Simulation
$V_{DS} = 18V; V_{GS} = 20V$	$\approx 1100A$	1125A
$V_{DS} = 11V; V_{GS} = 12V$	$\approx 800A$	743A
$V_{DS} = 20V; V_{GS} = 8V$	$\approx 540A$	551A
$V_{DS} = 9V; V_{GS} = 10V$	$\approx 600A$	574A
$V_{DS} = 19V; V_{GS} = 10V$	$\approx 60A$	873A
$V_{DS} = 1V; V_{GS} = 20V$	$\approx 100A$	98.7A

Table 1.3: output characteristics comparison @25°C

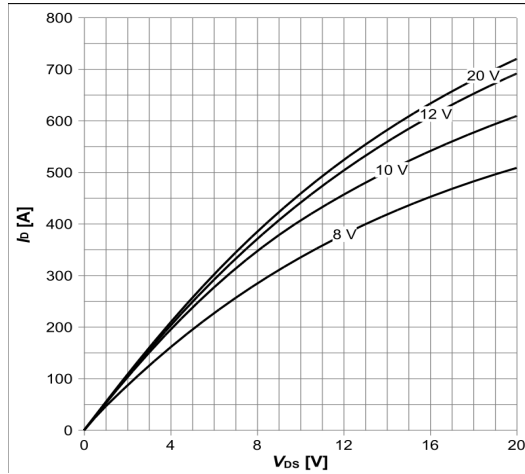


Figure 1.9: Datasheet output char @125°C

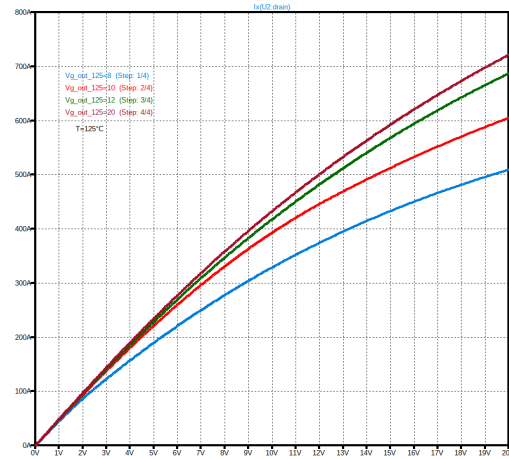


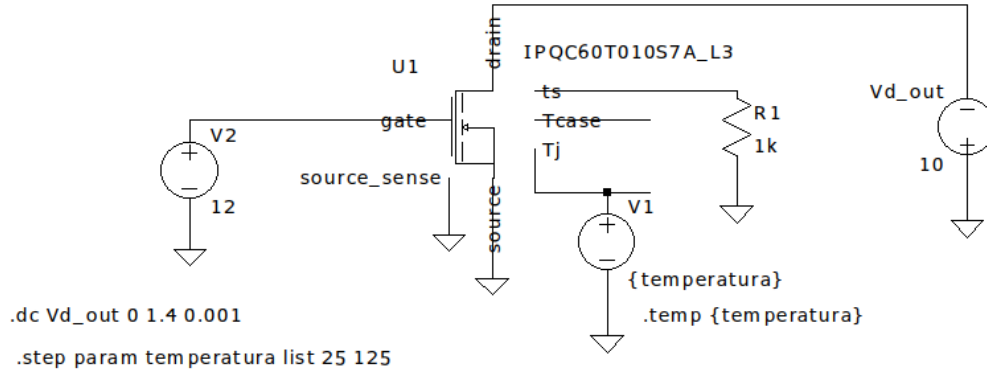
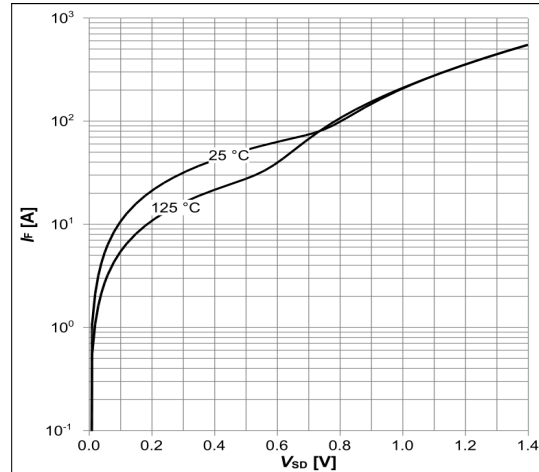
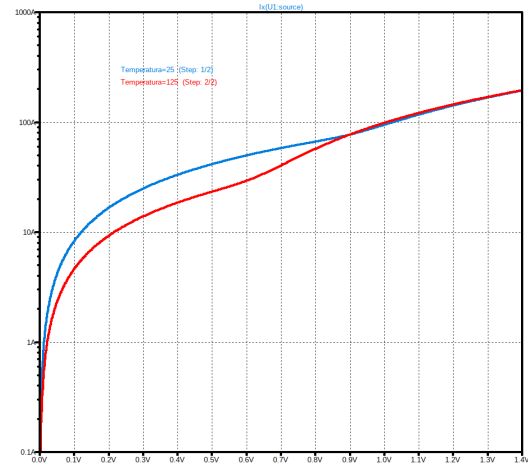
Figure 1.10: SPICE output char @125°C

$V_{DS}; V_{GS}$	I_D Datasheet	I_D Simulation
$V_{DS} = 1V; V_{GS} = 20V$	$\approx 50A$	48.7A
$V_{DS} = 3V; V_{GS} = 20V$	$\approx 150A$	147.6A
$V_{DS} = 19V; V_{GS} = 20V$	$\approx 700A$	698.7A
$V_{DS} = 19V; V_{GS} = 10V$	$\approx 600A$	587.4A
$V_{DS} = 16V; V_{GS} = 8V$	$\approx 450A$	448.9A

Table 1.4: output characteristics comparison @ 125°C

It must be noted that the model does not perfectly fit the behaviour of the real device, however since the application in study will never exceed 100A of drain current the model is still valid

Reverse diode characteristics


Figure 1.11: Schematic for Reverse diode validation

Figure 1.12: Datasheet Reverse diode

Figure 1.13: SPICE Reverse diode

$T_j; V_{DS}$	I_D Datasheet	I_D Simulation	Notes
$T_j = 25^\circ\text{C}; V_{DS} = 0.1\text{V}$	$\approx 10\text{A}$	8.98A	-
$T_j = 25^\circ\text{C}; V_{DS} = 0.2\text{V}$	$\approx 20\text{A}$	18.4A	-
$T_j = 125^\circ\text{C}; V_{DS} = 0.2\text{V}$	$\approx 10\text{A}$	9.87A	-
$T_j = 125^\circ\text{C}; V_{DS} = 0.4\text{V}$	$\approx 20\text{A}$	18.8A	-

Table 1.5: Reverse diode characteristics comparison

1.1.2 Dynamic characteristics

Typical gate charge

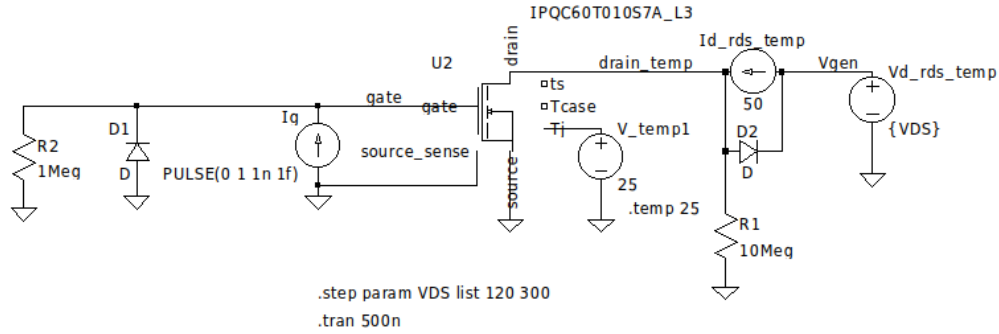


Figure 1.14: Schematic for typical gate charge

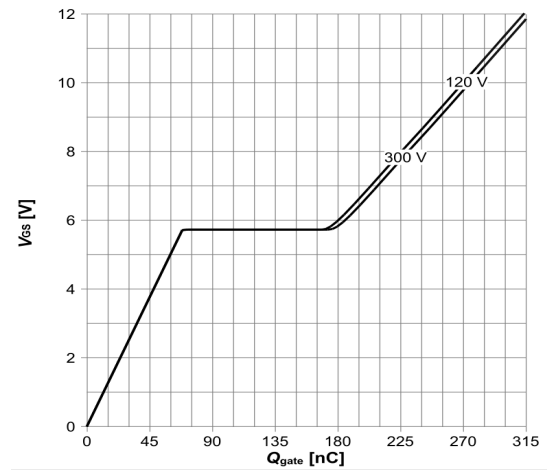


Figure 1.15: Datasheet gate charge

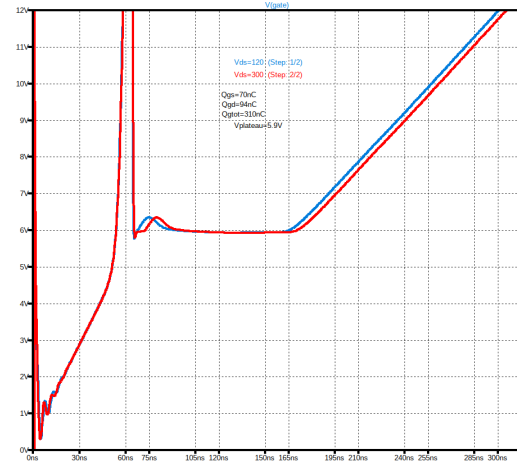


Figure 1.16: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=318nC	Qgtot=310nC
Qgs=69nC	Qgs=70nC
Qgd=105nC	Qgd=94nC
$V_{plateau} = 5.7V$	$V_{plateau} = 5.9V$

Table 1.6: Gate charge data comparision

Chapter 2

Performance improvements

From the simulations results it stands out that the main culprit of the entire system losses are in order

1. GaN transistors
2. Bulk capacitors
3. Silicon transistors

2.0.1 GaN transistor analysis

The main loss component of the GaN losses are the switching type, this is due to the implementation of hard-switching technique of the devices, which is proven by analyzing the waveform of the devices during a transition

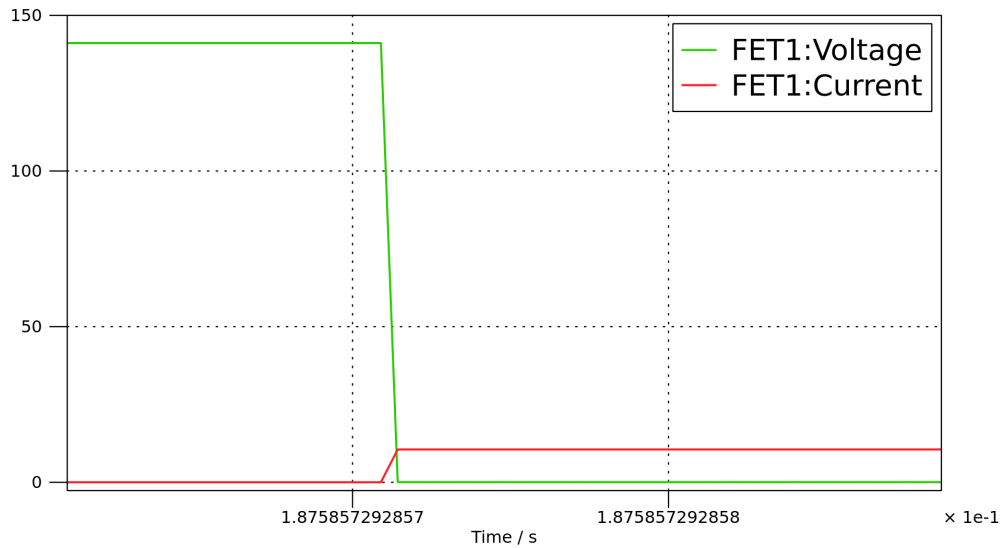


Figure 2.1: GaN turn on behaviour

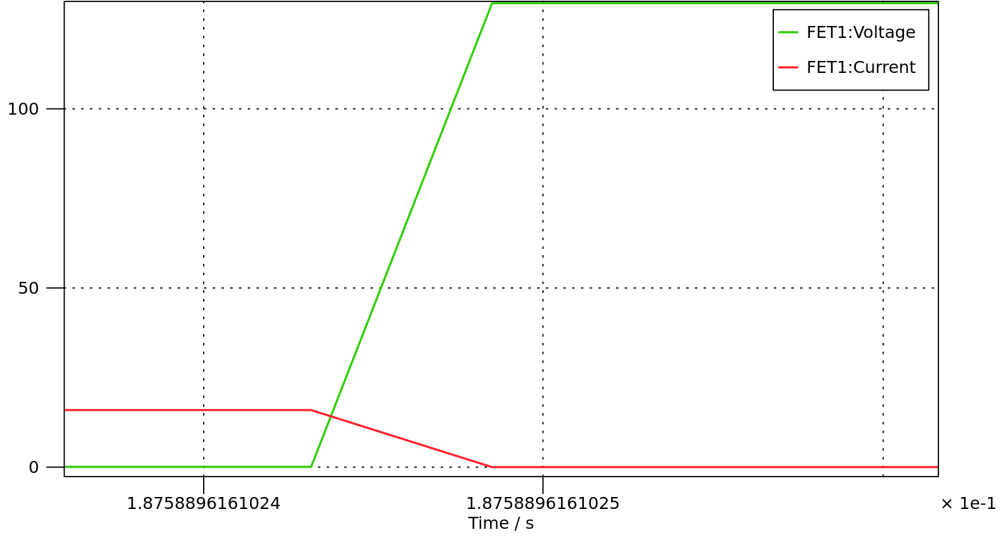


Figure 2.2: GaN turn off behaviour

the superimposition of the voltage and current applied on the device will create a very brief yet high power surge which leads to high switching losses. A soft switching technique ,such as ZVS ZCS,can be implemented in order to drastically reduce those losses contribution.

2.0.2 Bulk caps analysis

The bulk caps are aluminum electrolytic caps, by their nature those are lossy components, since are THT components , both ESR and ESL are high due to the nature of the solder joints. However due to the nature of the device, very high capacitance values are required to fulfill the needed duties of those components, by choosing a low ESR electrolytic capacitor with similar ratings losses can be reduced.

The choice fell upon the MAL219847331E3 [3] whose ESR is approximately $\frac{1}{3}$ of the Bulk capacitors currently installed, which will lead to a corresponding reduction in power losses

2.0.3 Silicon transistor analysis

Silicon transistors mainly experience conduction losses, so the best way to improve them is to find some components with similar voltage/current rating, but lower drain-source resistance. The choice fell upon the IPQC60T010S7A [2] which has a different package type but approximately $\frac{1}{2}$ of the current FET R_{dson} .

2.1 Performance comparison

As before the entire PFC system has been simulated with the same operating conditions as before in order to perform a fair comparison between the 2 systems.

The results are the following.

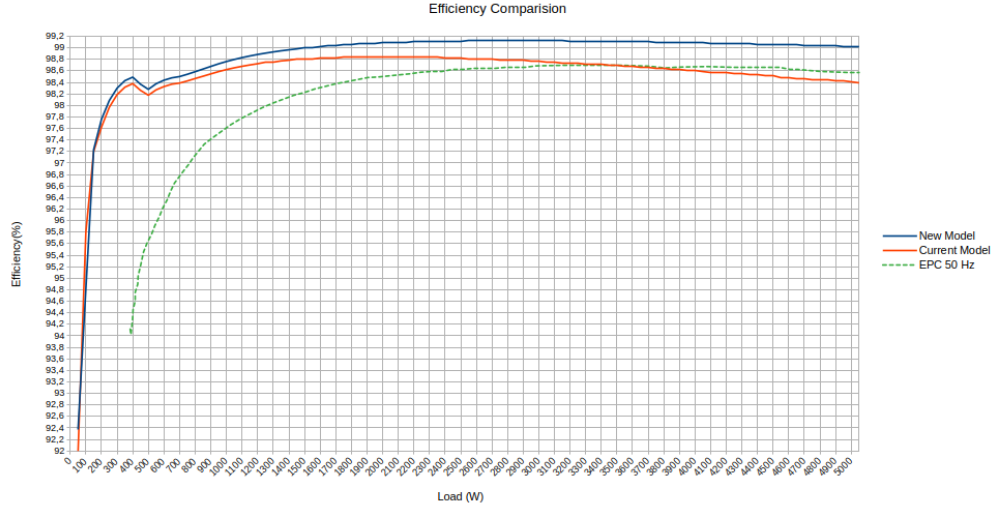


Figure 2.3: Efficiency comparison

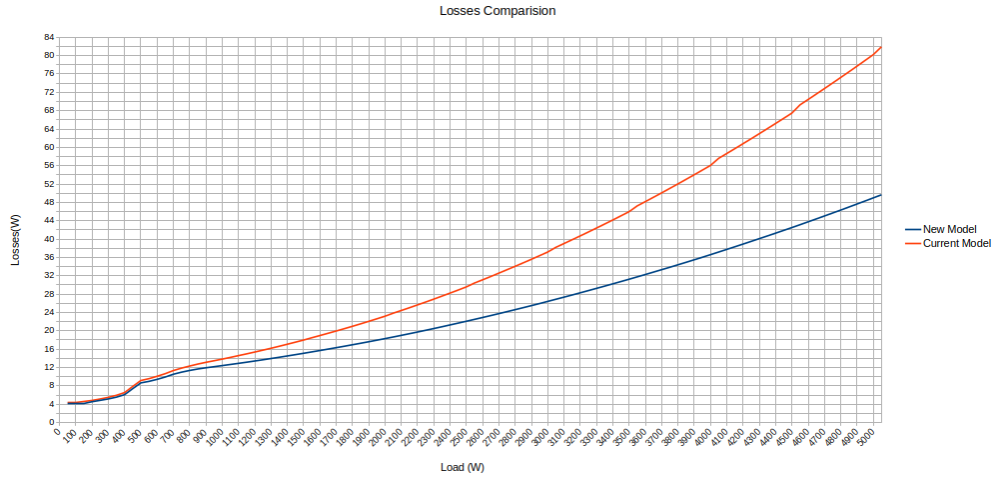


Figure 2.4: Losses comparison

It can be clearly seen that by introducing the 2 new components efficiency does in fact improve, losses went from 80W to 50W [2.4] by reducing the ohmic losses of both the Silicon FETs and the Bulk cap. However as we can see from [2.5] the main culprit are still the GaN devices with their switching losses [2.9]. This means that an effort to effectively implement a ZVS on those devices will drastically improve the performance of the system.

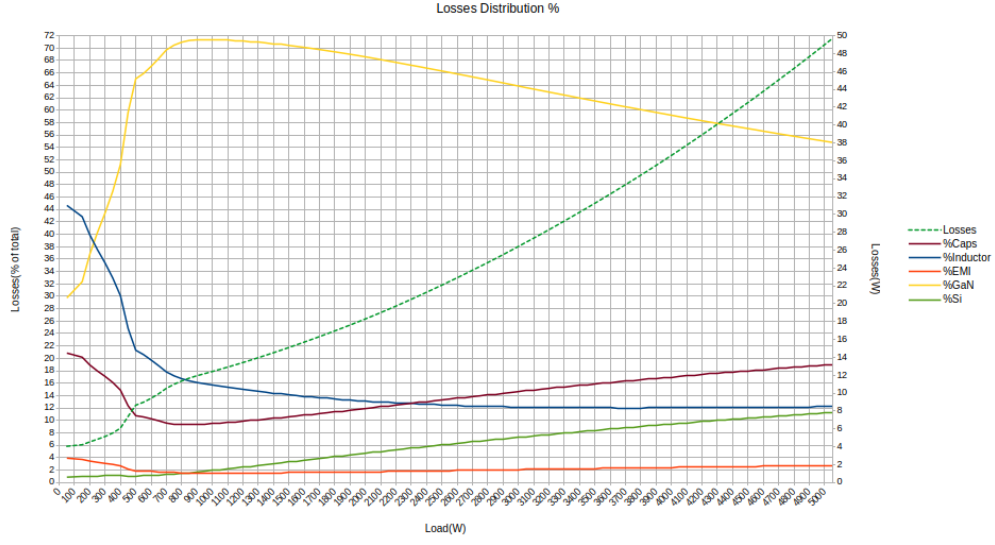


Figure 2.5: Losses distribution %

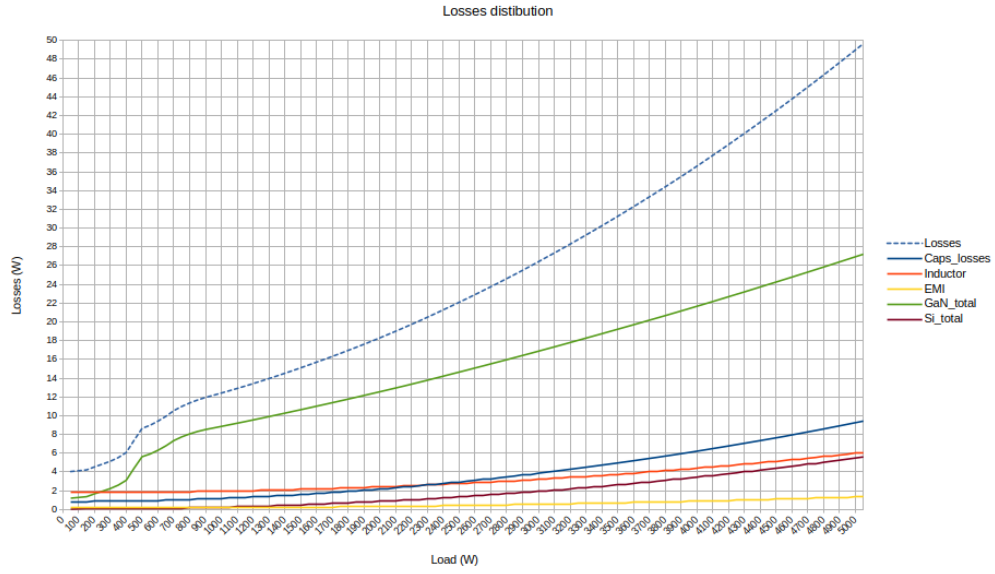


Figure 2.6: Losses distribution

The difference in losses of the 2 systems is highlighted in the following graphs , were only the switched components are isolated.

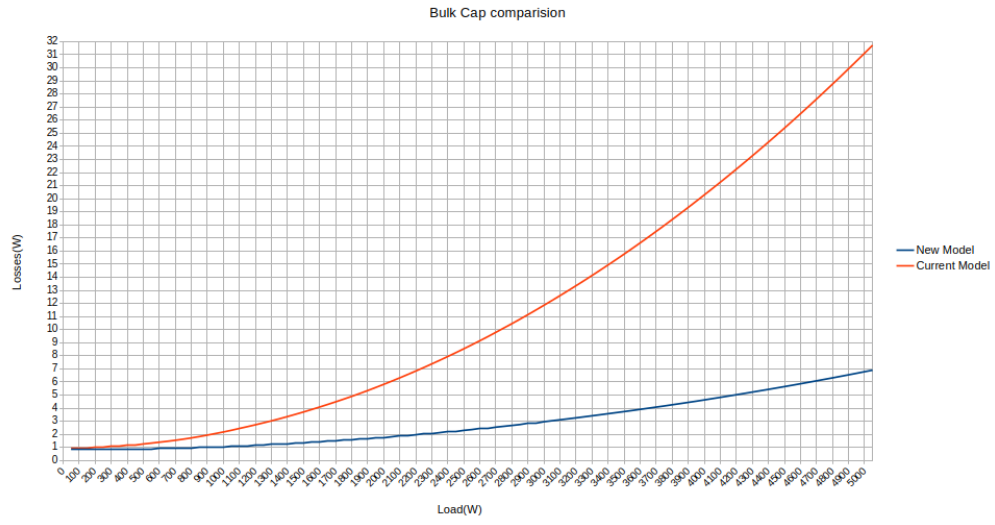


Figure 2.7: Bulk comparison

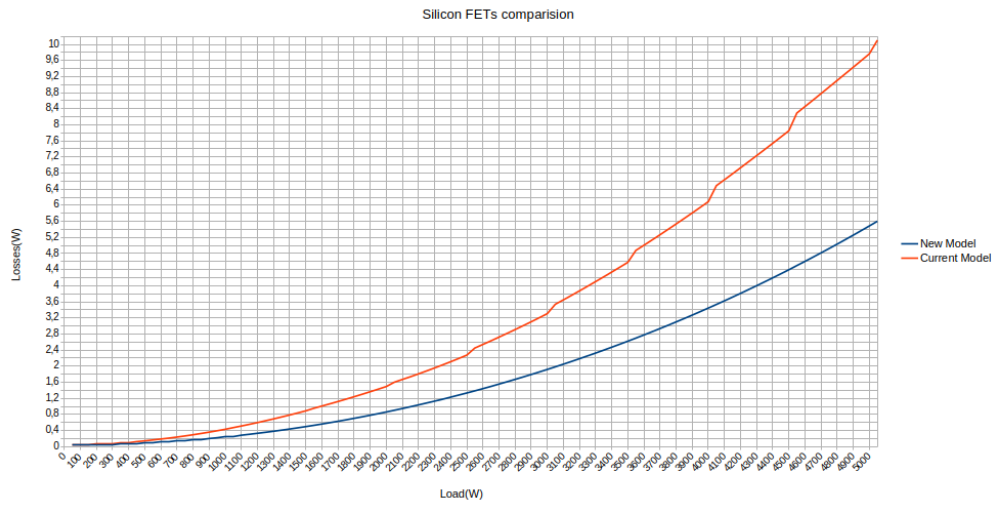


Figure 2.8: Si FETs comparison

A complete breakdown of the losses on all devices has been performed.

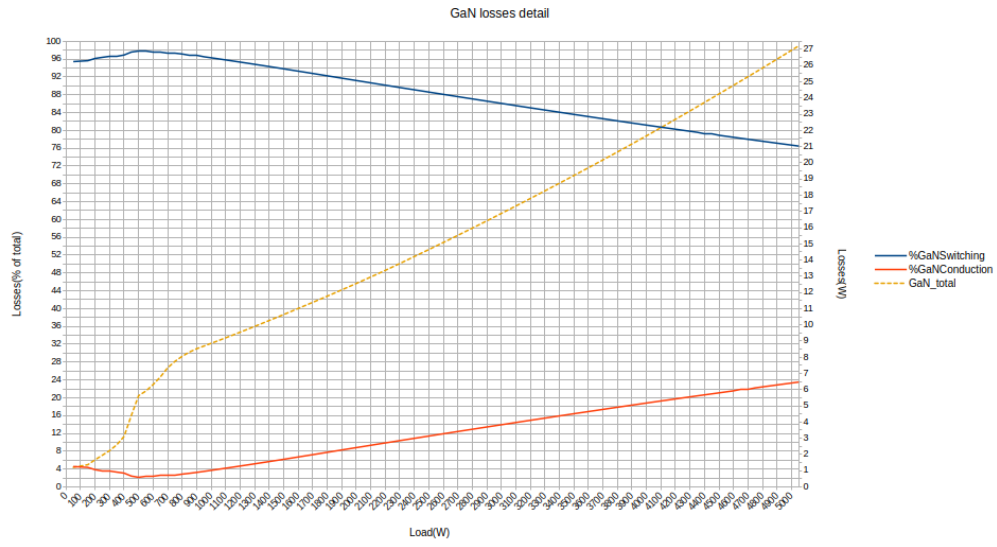


Figure 2.9: GaN losses detail

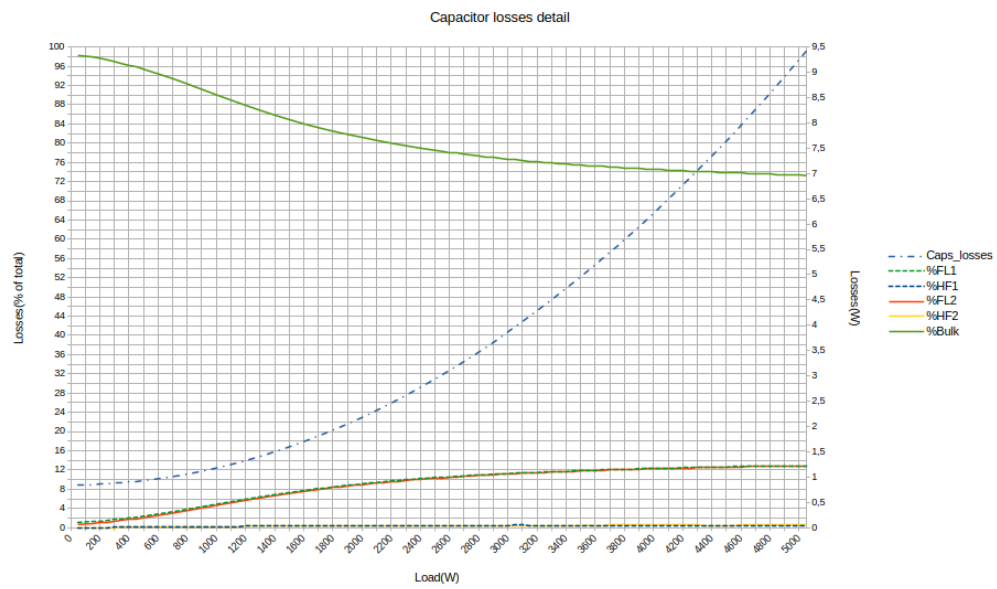


Figure 2.10: capacitor losses detail

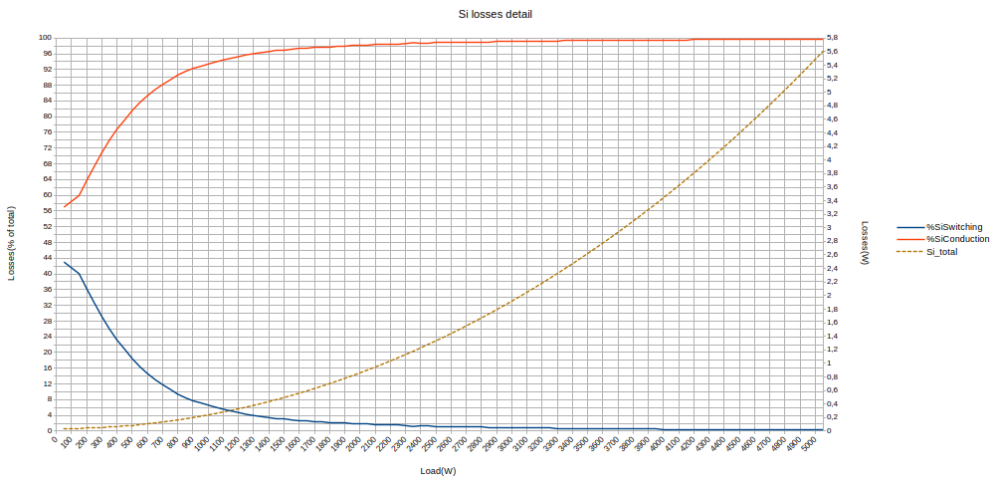


Figure 2.11: Si FET losses detail

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Bibliography

- [1] Giorgio Armignacco. *Thesis repository*. <https://github.com/arminioboy/PoliT0Thesis>. 2026 (cit. on p. 3).
- [2] Infineon. *IPQC60T010S7A*. <https://www.infineon.com/assets/row/public/documents/24/49/infineon-ipqc60t010s7a-datasheet-en.pdf>. 2024 (cit. on pp. 3, 12).
- [3] VISHAY. *Aluminum Electrolytic Capacitors, Power High Ripple Current Snap-In*. <https://www.vishay.com/docs/28339/198phr.pdf>. 2016 (cit. on p. 12).